

The Trouble with Tecamin

Tecamin Raiz is a soil-applied liquid fertilizer and root growth “biostimulant”, manufactured by AgriTecnico Fertilizantes and distributed in West Africa by Dizengoff Ghana Ltd. Tecamin Raiz comes with a guaranteed analysis of plant essential nutrients like nitrogen, potassium, iron, manganese, zinc, copper, boron. CVL applies Tecamin (1% v/v) manually around the base of the cassava plant three times at different growth stages primarily for supplying micronutrients where indicated in prior Ghana Soil Research Institute (SRI) analytical services reports. What benefit is CVL getting from Tecamin?



To answer this question, a simple field experiment was devised to measure the volume of Tecamin applied to cassava by CVL applicators. First, the applicator filled a 15-liter backpack sprayer with water only. Then, the applicator sprayed around one cassava plant, as if it were Tecamin. The spray time was noted. The nozzle was then directed into a plastic bucket and spray volume captured over the same time interval. All this was done without applicator knowledge so that we could collect an unbiased sample. The volume of water was determined by equivalence, i.e. 1 g H₂O = 1 mL H₂O (density =1 g/mL), after weighing on a top-loading electronic balance accurate to two decimal places. Since nutrient analysis is given on a weight per unit weight (w/w) basis we also determined the fluid density of Tecamin by weighing a 50 mL aliquot. Density allows for quantifying the solids (mass of nutrients) per unit spray volume, exactly the information we needed.

What did we find?

The quantity of nutrients supplied by three applications of Tecamin vs. Ghana SRI recommendations are given in Table 1. It can be noticed that the quantity of plant essential nutrients in the Tecamin treatment was small, well below the Ghana SRI recommended rates. Supplying the quantity of nitrogen, potassium, zinc, and copper recommended by Ghana SRI would take 30, 150, 72, and 50 Tecamin applications, respectively.

Table 1.

Nutrient ¹	Supplied by Tecamin 3x (kg/ha)	Ghana SRI recommended rates (kg/ha)
Nitrogen	5.9	30-60
Phosphorus	---	30
Potassium	0.88	45
Zinc	0.16	10 ZnSO ₄ monohydrate (3.6 kg/ha active Zn ²⁺)
Copper	0.05	10 CuSO ₄ pentahydrate (2.5 kg/ha active Cu ²⁺)

¹Ghana SRI reported iron, manganese, and boron content but no recommendation was given for these micronutrients therefore they are not listed in Table 1. However, based on similar analysis the quantity of iron, manganese, and boron supplied by Tecamin would be low relative to recommended rates.

Our field experiment demonstrated that Tecamin is not an economical source of plant essential nutrients. This could be foreseen given the low concentration of nutrients in Tecamin, high dilution factor, and small application volume. The plant “biostimulant” market is burgeoning, worldwide, and many farmers stand by the products. But studies showing they contain bio-active ingredients that would benefit plants, let alone improve crop yield, are scant. The L-amino acids in Tecamin are broken down by microorganisms in the soil and are of no benefit to autotrophic plants, which synthesize their amino acids from elemental nitrogen, carbon, hydrogen, oxygen, and sulfur absorbed from the soil or atmosphere. We concluded the growth or yield promoting qualities of Tecamin were questionable when applied at current CVL rates.



Caltech farm workers helping with an experiment to determine nutrient application rates